

CME 307 / MS&E 311 / OIT 676: Optimization

LP geometry, modeling and solution techniques

Professor Udell

Management Science and Engineering
Stanford

September 24, 2026

First quiz Thursday next week

- ▶ Take paper when you arrive
- ▶ Write your SUNetID¹ and your name.
- ▶ We will project the question.
- ▶ You will have 5 minutes to write your answer on the paper.
- ▶ When time is up, put your pen or pencil down and stop writing.

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since we are calibrating what is possible with a 5 minute quiz, we will grade on a **check** basis.

- ▶ **no check**: no serious attempt
- ▶ **check**: a serious attempt
- ▶ **check+**: correct or nearly correct

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Exit ticket

Please fill out the exit ticket (linked from course schedule spreadsheet) before you leave:

- ▶ Helps us improve the course

We will not grade this, but we will read it!



Course overview

the practice of optimization consists of:

- ▶ modeling (real-world problems $\rightarrow$ optimization problems)
- ▶ analysis (properties of optimization problems)
- ▶ algorithms (how to solve optimization problems)

all three are necessary for success in optimization!

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- ▶ last Monday: mostly modeling
- ▶ today: mostly analysis
- ▶ next Monday: algorithms and some more modeling

Outline

LP standard form

LP inequality form

What kinds of points can be optimal?

Solving LPs

Modeling

Linear programming: standard form

standard form linear program (LP)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & x \geq 0\end{array}$$

optimal value p^* , solution x^* (if it exists)

- ▶ any x with $Ax = b$ and $x \geq 0$ is called a **feasible point**
- ▶ if problem is infeasible, we say $p^* = \infty$
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Q: why? how to check?

A: otherwise infeasible or redundant rows; use gaussian elimination to check and remove

LP example: diet problem

We are planning a backpacking trip, and want to minimize the total weight of the food packed subject to nutritional requirements. We have a list of essential nutrients and how much an active person needs of each. We also know the weight of each food, and how much of each nutrient is in each food.

- ▶ x_j servings of food j , $j = 1, \dots, n$
- ▶ c_j weight per serving
- ▶ a_{ij} amount of nutrient i in food j
- ▶ b_i required amount of nutrient i , $i = 1, \dots, m$

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the **feasible set** is the set of points $\{x \mid Ax = b, x \geq 0\}$ that satisfy all constraints.

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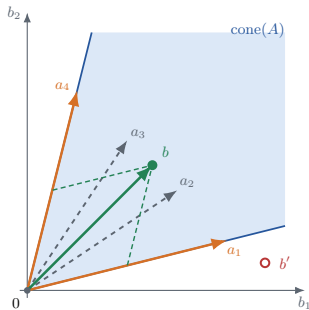
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A: at most m (here, b uses only a_1 and a_4)



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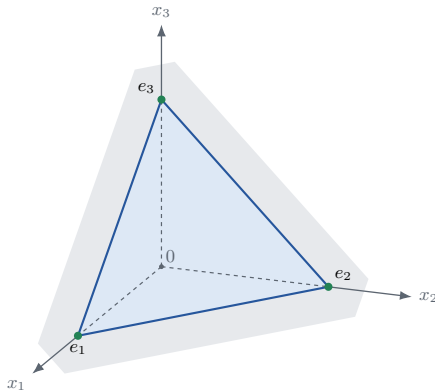
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- ▶ LP is feasible if hyperplane $\{x \mid Ax = b\}$ intersects the positive orthant

example: in $\mathbb{R}^3$, $\mathbf{1}^T x = 1$ cuts the orthant in the triangle $\text{conv}\{e_1, e_2, e_3\}$



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$$\text{conv}(S) = \left\{ \sum_{i=1}^k \theta_i x_i \mid x_i \in S, \theta_i \geq 0, \sum_{i=1}^k \theta_i = 1 \right\}$$

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- ▶ the feasible set $\{x : Ax = b, x \geq 0\}$ is convex

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LP inequality form

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interpretation: halfspaces

- ▶ $a_i^T x \leq b_i$ defines a **halfspace**
- ▶ $Ax \leq b$ defines a **polyhedron**: intersection of halfspaces
- ▶ LP is feasible if polyhedron $\{x \mid Ax \leq b\}$ is nonempty

LP example: production planning

- ▶ x_i units of product i
- ▶ c_i cost per unit
- ▶ a_{ij} amount of resource j used by product i
- ▶ b_j amount of resource j available
- ▶ d_i demand for product i

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 $c^T x + f^T z$, $z_i \in \{0, 1\}$, $x_i \leq Mz_i$ for M large

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$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \leq b \end{array} \quad \rightarrow \quad \begin{array}{ll} \text{minimize} & c^T (x_+ - x_-) \\ \text{subject to} & A(x_+ - x_-) + s = b \\ & s, x_+, x_- \geq 0 \end{array}$$

so both forms have the same expressive power, and feasible sets are polyhedra

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active variables. for nonnegative variable $x \geq 0$, variable i is **active** if $x_i > 0$

example: active slack variables are dual to active constraints

$$Ax \leq b \iff Ax + s = b, s \geq 0$$

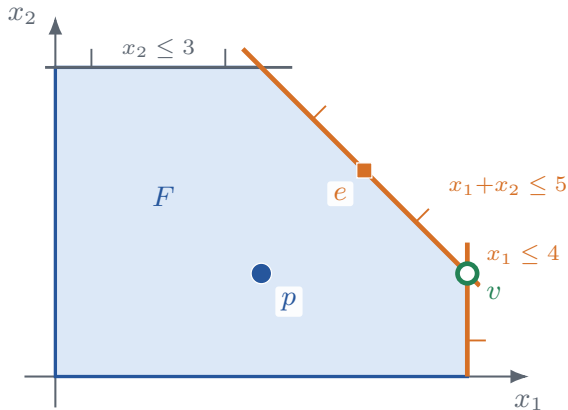
$$a_i^T x = b_i \iff s_i = 0$$

constraint i is active $\iff$ slack variable s_i is inactive

Active constraints: example

$$F = \left\{ x \in \mathbb{R}^2 \mid \begin{array}{l} x \geq 0, \ x_1 \leq 4, \\ x_2 \leq 3, \ x_1 + x_2 \leq 5 \end{array} \right\}$$

- ▶ corner $v = (4, 1)$: $x_1 \leq 4$ and $x_1 + x_2 \leq 5$ active
- ▶ edge point $e = (3, 2)$: only $x_1 + x_2 \leq 5$ active
- ▶ interior point $p = (2, 1)$: none active



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Q: Does there always exist an extreme solution?

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Vertices

define **vertex**: $x \in \mathbb{R}^n$ is a vertex of set $S \subset \mathbb{R}^n$ if for some vector $c \in \mathbb{R}^n$,

$$c^T x < c^T y \quad \forall y \in S \setminus \{x\}$$

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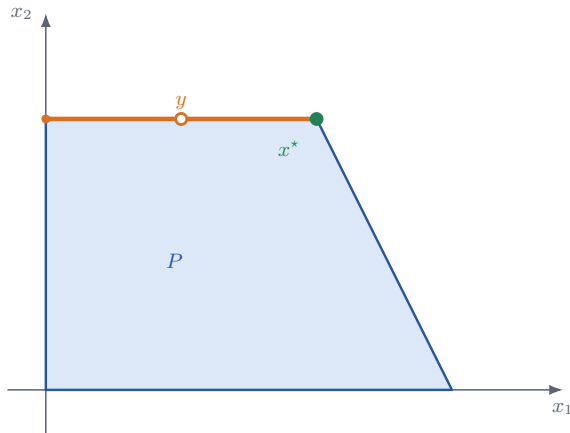
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x is the unique optimum of this problem, so the proof of this statement follows from the previous proof.

Extreme points and supporting hyperplanes

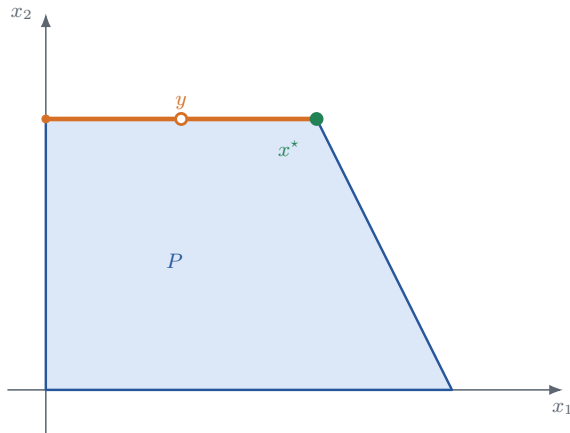
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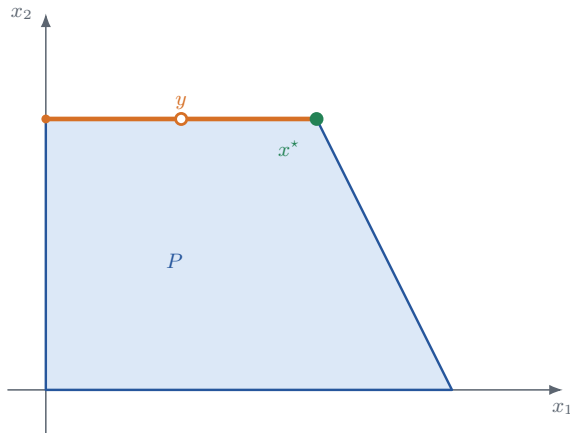
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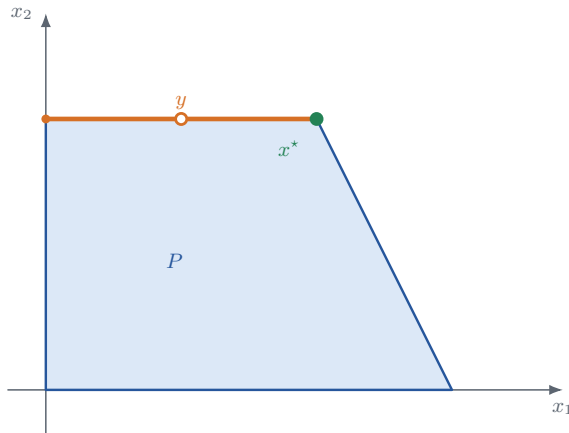
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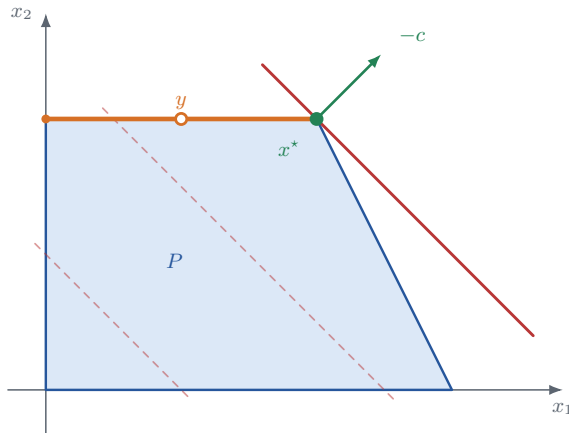
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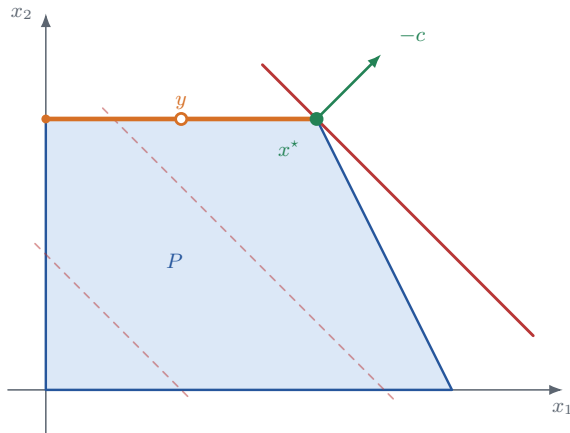
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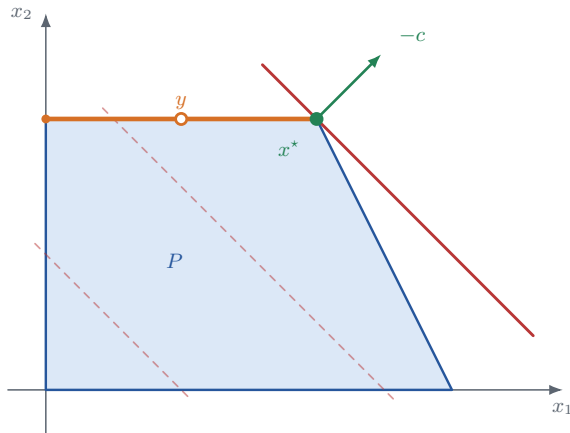
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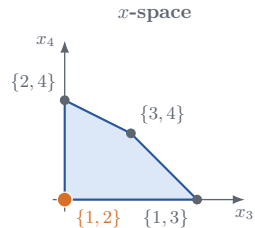
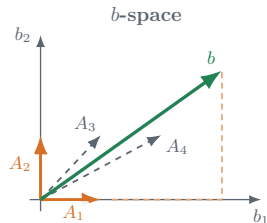
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A: choose m linearly independent columns of A and set $x = A_S^{-1}b$; check $x \geq 0$.

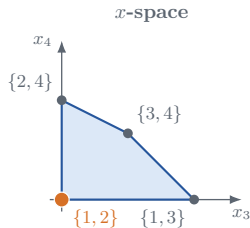
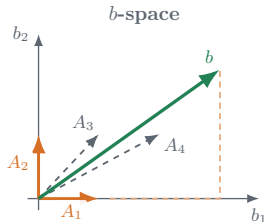
A basis produces a basic feasible solution

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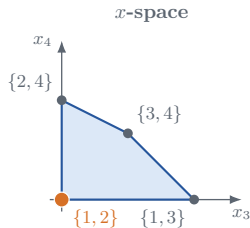
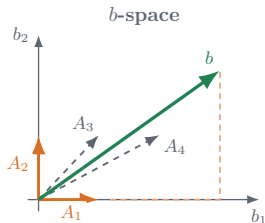
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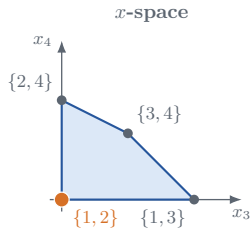
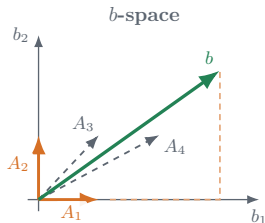


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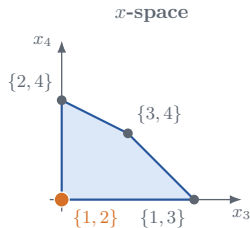
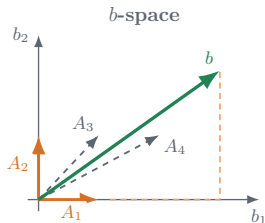
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A: a basis, but $x = (-1, 0, 0, 2) \not\geq 0$: basic but **infeasible**

Extreme point $\iff$ vertex $\iff$ BFS

fact. consider the feasible set in standard form $F = \{x \mid Ax = b, x \geq 0\}$ in $\mathbb{R}^n$.
the following are equivalent:

- ▶ x is an extreme point of F
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implications: since any polyhedron $Ax \leq b$ can be rewritten in standard form,

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we have already shown that vertex $\implies$ extreme point. need to show

- ▶ extreme point $\implies$ BFS
- ▶ BFS $\implies$ vertex

Extreme point $\implies$ BFS

let $x^\star$ be an extreme point of F , and let

$$S = \{i : x_i^\star > 0\}.$$

Extreme point $\implies$ BFS

let x^* be an extreme point of F , and let

$$S = \{i : x_i^* > 0\}.$$

1. The columns in A_S are linearly independent. Otherwise, there is a nonzero $d \in \mathbb{R}^n$, supported on S , with $Ad = 0$. Since $x_i^* > 0$ on S , for sufficiently small $\epsilon > 0$,

$$x^+ = x^* + \epsilon d, \quad x^- = x^* - \epsilon d$$

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contradicting that x^* is extreme.

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2. Complete A_S to a basis. Since A has rank m , extend S to B with $|B| = m$ and A_B invertible. Then $x_B^* = 0$, and $Ax^* = b$ gives

$$x_B^* = A_B^{-1}b.$$

Thus x^* is a BFS.

BFS $\Rightarrow$ vertex

suppose x^* is a BFS of F with active set S and A_S invertible. define $c \in \mathbb{R}^n$ as

$$c_i = \begin{cases} 0 & \text{if } i \in S \\ 1 & \text{otherwise} \end{cases}$$

so $c^T x^* = 0$.

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- ▶ x^* is the only point in F supported on S , as $\text{nullspace}(A_S) = 0$,
- ▶ so any other feasible point $x \in F$ has a positive objective value $c^T x > 0$.

hence x^* is a vertex of F with defining vector c .

Outline

LP standard form

LP inequality form

What kinds of points can be optimal?

Solving LPs

Modeling

Solving LPs

algorithms:

- ▶ enumerate all vertices and check
- ▶ fourier-motzkin elimination
- ▶ simplex method
- ▶ ellipsoid method
- ▶ interior point methods
- ▶ first-order methods
- ▶ ...

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remarks:

- ▶ enumeration and elimination are simple but not practical
- ▶ simplex was the first practical algorithm; still used today
- ▶ ellipsoid method is the first polynomial-time algorithm; not practical
- ▶ interior point methods are polynomial-time and practical
- ▶ first-order methods are practical and scale to large problems

Example of Fourier-Motzkin elimination

consider the system of inequalities

$$x_1 + 2x_2 \leq 4$$

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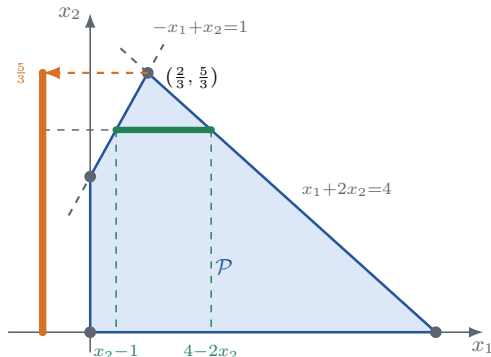
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elimination method also shows projection of a polyhedron is a (closed) polyhedron



Enumerate vertices of LP

can generate all extreme points of LP: for each $S \subseteq \{1, \dots, n\}$ with $|S| = m$,

- ▶ $A_S \in \mathbb{R}^{m \times m}$, submatrix of A with columns in S , is invertible
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- ▶ evaluate objective $c^T x$

the best BFS is optimal!

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n choose m is $\binom{n}{m} = \frac{n!}{m!(n-m)!}$ (“exponentially many”)

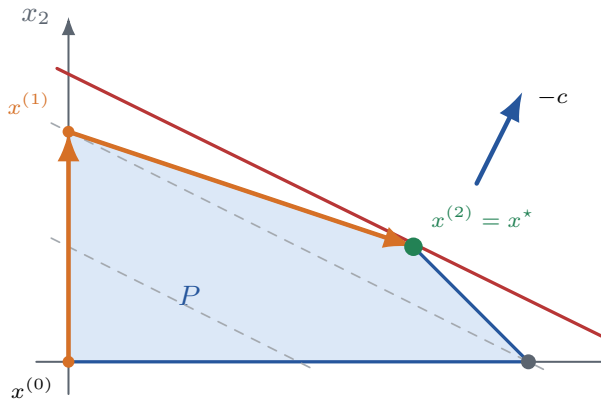
Simplex algorithm: example

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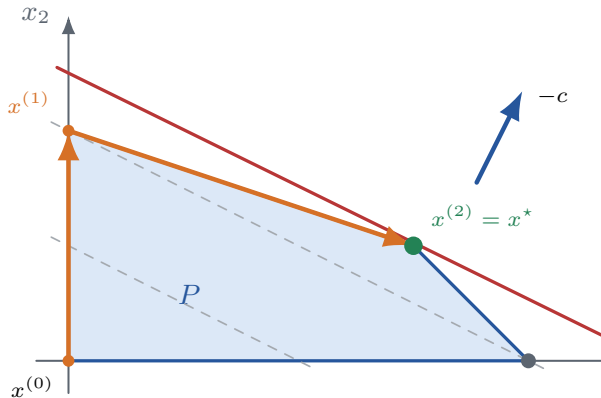
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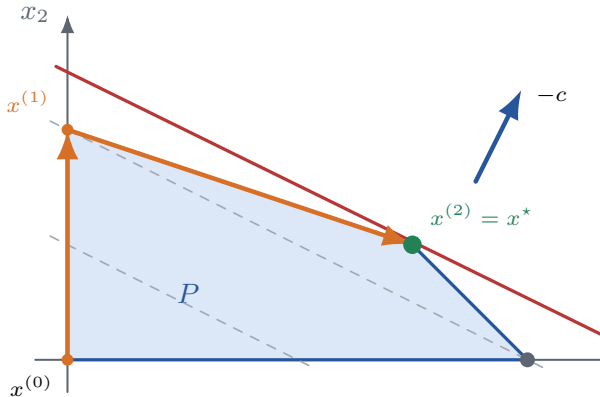
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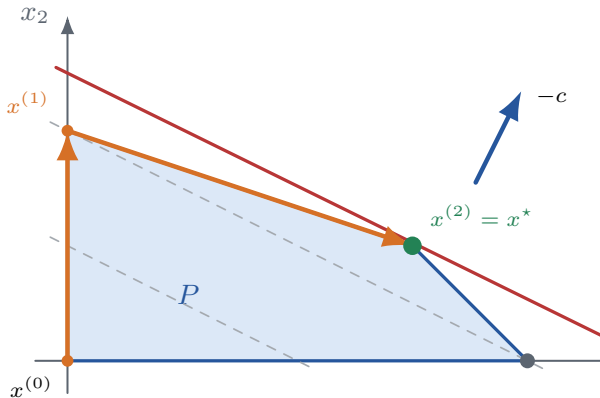
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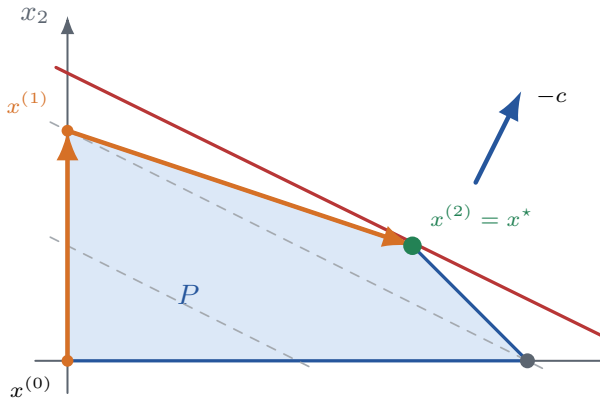


Simplex algorithm: example

$$\begin{array}{ll}\text{minimize} & -x_1 - 2x_2 \\ \text{subject to} & x_1 + x_2 \leq 4 \\ & x_1 + 3x_2 \leq 6 \\ & x \geq 0\end{array}$$

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- ▶ increase x_2 until a constraint binds:
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Q: why does x_2 stop at 2, not 4?



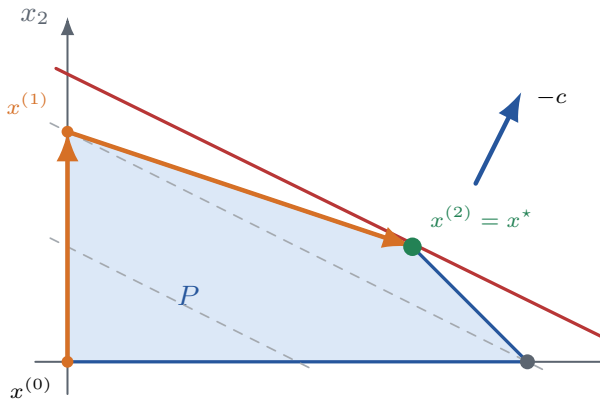
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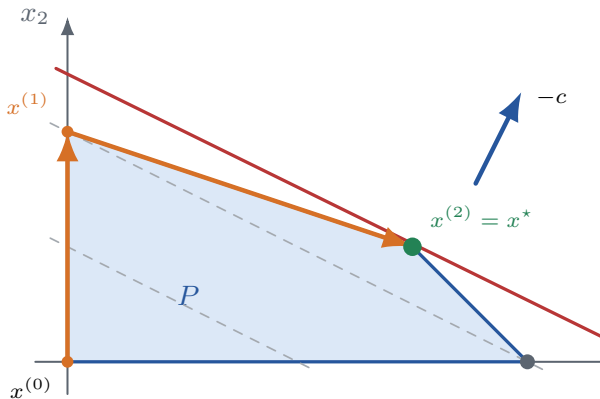
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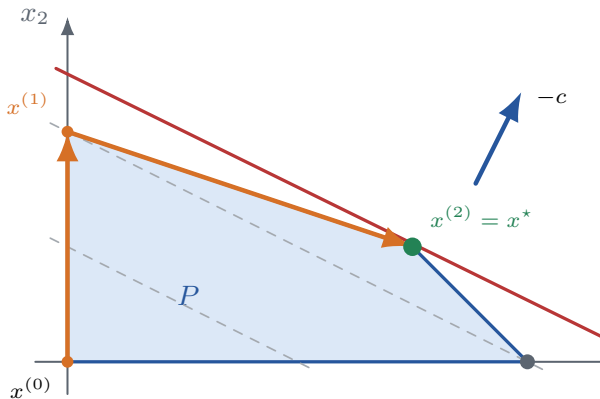
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Q: why stop at $x^{(2)}$?

A: no neighboring vertex improves the objective



Simplex algorithm

basic idea: local search on the vertices of the feasible set

- ▶ start at BFS x and evaluate objective $c^T x$
- ▶ move to a neighboring BFS x' with better objective $c^T x'$
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discuss in groups:

- ▶ how to find an initial BFS?
- ▶ how to find a neighboring BFS with better objective?
- ▶ how to prove optimality?

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- ▶ $x = 0, z = |b|$ is a BFS of this problem
- ▶ $(x, z) = (x, 0)$ is a BFS of this problem $\iff x$ is a BFS of the original problem

Find a better neighboring BFS

start with BFS x with active set S , $x_S > 0$. (called a **non-degenerate** BFS.)
construct the **j th basic direction** d^j by turning on variable $j \notin S$

$$x^+ \leftarrow x + \theta d^j, \quad \theta > 0$$

where $d_j^j = 1$ and $d_i^j = 0$ for $i \notin S \cup \{j\}$. need to solve for d_S^j .

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- ▶ as $x_S > 0$ is non-degenerate, $\exists \theta > 0$ st $x^+ \geq 0$
- ▶ how does objective change if we move to $x^+ = x + \theta d^j$?

$$c^T x^+ - c^T x = \theta c^T d^j = \theta c_j - \theta c_S^T A_S^{-1} A_j$$

Reduced cost

define **reduced cost** $\bar{c}_j = c_j - c_S^T A_S^{-1} A_j, j \notin S$

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we will show:

- ▶ if $\bar{c} \geq 0$, x is optimal
- ▶ if x is optimal and nondegenerate ($x_S > 0$), then $\bar{c} \geq 0$

(nondegeneracy is needed only for the converse)

if $\bar{c} \geq 0$, x is optimal

let x be a BFS with basis S , and let x' be any feasible point. Since

$$A_S x_S = b, \quad A_S x'_S + A_{\bar{S}} x'_{\bar{S}} = b,$$

we have

$$x'_S = x_S - A_S^{-1} A_{\bar{S}} x'_{\bar{S}}.$$

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substitute into the objective:

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because $\bar{c}_{\bar{S}} \geq 0$ and $x'_{\bar{S}} \geq 0$. therefore no feasible point has lower objective value than x , so x is optimal.

Outline

LP standard form

LP inequality form

What kinds of points can be optimal?

Solving LPs

Modeling

LP solvers

practical solvers for MILP:

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https://jump.dev/JuMP.jl/stable/tutorials/applications/power_systems/

Oro Verde case + tutorial

<https://github.com/stanford-cme-307/demos/tree/main/gurobipy>

Modeling challenges

model the following as standard form LPs:

1. **inequality constraints.** $Ax \leq b$
2. **free variable.** $x \in \mathbb{R}$
3. **absolute value.** constraint $|x| \leq 10$
4. **piecewise linear.** objective $\max(x_1, x_2)$
5. **assignment.** e.g., every class is assigned exactly one classroom
6. **logic.** e.g., class enrollment $\leq$ capacity of assigned room
7. **(big-M).** $Ax \leq b$ if $x \geq 10$
8. **flow.** e.g., the least cost way to ship an item from s to t

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(see chapter 1 of Bertsimas and Tsitsiklis for more details on 1–6. see

[https://github.com/stanford-cme-307/demos/blob/main/](https://github.com/stanford-cme-307/demos/blob/main/Multicast_Routing_Demonstration.ipynb)

[Multicast_Routing_Demonstration.ipynb](https://github.com/stanford-cme-307/demos/blob/main/Multicast_Routing_Demonstration.ipynb) for a detailed treatment of a flow problem.)

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introduce slack variable $s \in \mathbb{R}^m$: $Ax + s = b, s \geq 0 \iff Ax \leq b$

$$\begin{array}{ll}\text{minimize} & c^T x + 0^T s \\ \text{subject to} & Ax + s = b \\ & x, s \geq 0\end{array}$$

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introduce positive variables x_+, x_- so $x = x_+ - x_-$:

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Q: Why does this work? For what kinds of functions can we use this trick?

Use binary variables to handle assignment

every class is assigned exactly one classroom:

define variable $X_{ij} \in \{0, 1\}$ for each class $i = 1, \dots, n$ and room $j = 1, \dots, m$

$$X_{ij} = \begin{cases} 1 & \text{class } i \text{ is assigned to room } j \\ 0 & \text{otherwise} \end{cases}$$

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now solve the problem

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^n \sum_{j=1}^m C_{ij} X_{ij} \\ \text{subject to} & \sum_{j=1}^m X_{ij} = 1, \forall i \quad (\text{every class assigned one room}) \\ & \sum_{i=1}^n X_{ij} \leq 1, \forall j \quad (\text{no more than one class per room}) \\ & X_{ij} \in \{0, 1\} \quad (\text{binary variables}) \end{array}$$

where C_{ij} is the cost of assigning class i to room j .

Use binary variables to handle logic

model class enrollment $p_i \leq$ capacity c_j of assigned room:

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what if we want enrollment p to be a variable, too?

...or use a big-M relaxation!

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suppose M is a very large number. solve the problem

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